DIGITAL SIGNAL PROCESSING · LABORATORY PROJECT

Detection of Synthetic Images by Masked Spectrum Similarity Analysis

Input–Output Results Report

Measured output of the implemented MaskSim detector on a nine-image test set

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METHOD	MaskSim — masked spectrum similarity analysis (Li, Bammey, Gardella, Nikoukhah, Morel, Colom & Grompone von Gioi, CVPR Workshops 2024, pp. 3855–3865)
DETECTORS EVALUATED	7 generator-specific models + 1 generic (maximum) model
TEST INPUTS	9 images — 3 photographs, 5 synthetic, 1 of unverified provenance
CALIBRATION POPULATION	18,814 known-real photographs (COCO-unlabeled-2017 + FlickrFaces)
COMPUTE	CPU inference, 19.7 s mean per image across 7 detectors

1 Purpose and scope

This report records the **measured input and output** of the synthetic-image detector built for this project. For each of nine test images it states precisely what was submitted to the system and what the system returned: the raw similarity score from each of the seven generator-specific detectors, the calibrated false-positive rate derived from each score, and the final decision of the generic model.

Every number in this document was produced by executing the implemented pipeline on the images listed in Section 3. No value has been estimated, interpolated or taken from the source publication. Where the system was wrong, the error is reported as it occurred.

> **What this report is not.** Nine images cannot establish detection accuracy. The figures in Section 7 describe the outcome of this specific run and are reported as a demonstration of system behaviour, not as a statistical evaluation. A proper evaluation would require the full Synthbuster benchmark and is identified as outstanding work in Section 10.

2 System under test

The detector is the MaskSim architecture: a frozen DnCNN residual extractor suppresses image content, the two-dimensional DFT of the residual is taken, its log-magnitude is weighted by a learned frequency mask, and the masked spectrum is compared by cosine similarity against a learned reference spectrum. The resulting score is converted to a false-positive rate by ranking it against a large population of known-real photographs.

Detector architecture	Frozen DnCNN (17 layers, 3×3, residual) → 2-D DFT → log-magnitude → learned mask → cosine similarity vs. learned reference → logistic head
Trainable parameters per detector	1,371,869 (mask 685,926 · reference 685,926 · 1×1 convolution 9 · batch-norm 6 · logistic 2)
Frozen parameters	≈558,000 (DnCNN, shared and never fine-tuned)
Checkpoints used	`checkpoints/JPEG_Qrandom_w512/newsynth/` — seven models trained with random-quality JPEG augmentation at 512 px
Analysis window	512 × 512 px centre crop; 478 × 478 valid region after the DnCNN border
Calibration data	18,814 scores per detector, from `calibration/coco_unlabled_2017_to_*.csv` and `calibration/FlickrFace_to_*.csv`
Generic model	Maximum of the seven *calibrated* scores, re-calibrated against `calibration/all_to_max.csv` to correct for multiple comparisons
Execution	CPU, PyTorch; mean 19.7 s per image for all seven detectors

3 Test inputs

Nine images were submitted. Provenance is stated as *declared*: it comes from the file name, the originating repository, or embedded camera metadata, and is recorded here exactly as strongly as the evidence supports.

FILE	DECLARED PROVENANCE	FORMAT	NATIVE SIZE	BYTES
`real_img.png`	Photograph (repository reference)	PNG	512 × 512	399,509
`fake_img.png`	Synthetic (repository reference)	PNG	512 × 512	587,156
`kitten_real.jpg`	Photograph (EXIF-verified)	MPO	3336 × 3000	3,590,426
`mountain_real.jpg`	Photograph (asserted)	JPEG	2732 × 2732	1,006,732
`cat.jpg`	Unverified	JPEG	1370 × 1148	494,232
`Adobe_Firefly_…_photography.jpg`	Synthetic (Adobe Firefly)	JPEG	960 × 960	291,505
`sdxl_text.jpg`	Synthetic (Stable Diffusion XL)	JPEG	1024 × 1024	376,920
`stable-diffusion-demo-1-1.webp`	Synthetic (Stable Diffusion 1.x)	WEBP	1000 × 738	96,212
`Astronaut_Riding_a_Horse_(SD3.5).webp`	Synthetic (Stable Diffusion 3.5)	WEBP	1024 × 1024	88,204

Two of the three images presented as photographs carry no camera metadata; only `kitten_real.jpg` retains an EXIF record (Panasonic DMC-TS3, 23 May 2012). `cat.jpg` carries neither metadata nor an operator label and is therefore treated throughout as **provenance unverified**.

4 How to read the output

The system emits three quantities per detector. They are not interchangeable and are reported separately.

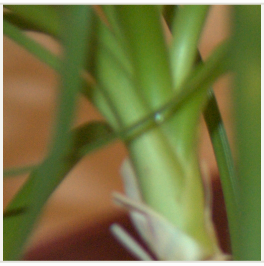
Raw cosine similarity	The cosine between the masked spectrum of the input and the detector's learned reference, in [0, 1]. It has no absolute meaning: its scale differs between detectors and it cannot be compared across them.
False-positive rate (FPR)	The fraction of the 18,814 known-real photographs in the calibration set that scored *at least as high* as this input. An FPR of 3% states that 3% of genuine photographs would be wrongly accused at this threshold. This is an empirical rank, not a learned output, and it is what makes the seven detectors comparable.
Interpretation band	FPR < 1% — very likely synthetic; < 5% — likely synthetic; < 10% — possibly synthetic; ≥ 10% — no evidence.

> **“No evidence” is not a clean bill of health.** It means the input did not resemble any of the seven learned fingerprints. A genuine photograph produces that result; so does a synthetic image from a generator the system was never trained on, and so does a synthetic image that has been resized or re-compressed enough to displace its spectral peaks.

5 Results, input by input

Each card below pairs one input with the complete output the system returned for it. The detector with the lowest false-positive rate is highlighted.

Input 1 · `real_img.png`



Declared provenance	Photograph (repository reference)
Container format	PNG
Native resolution	512 × 512 px
File size	390 KB
Analysed region	512 × 512 px centre crop
Runtime	21.6 s (cpu)

Macro photograph of plant stems, supplied with the MaskSim reference implementation as its canonical negative example.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00049	43.63%		no evidence
Stable Diffusion 2	0.00125	67.19%		no evidence
Stable Diffusion XL	0.00354	50.74%		no evidence
DALL·E 2	0.00311	73.50%		no evidence
DALL·E 3	0.00903	77.30%		no evidence
Midjourney	0.03553	48.79%		no evidence
Adobe Firefly	0.21147	13.87%	◀ lowest	no evidence

GENERIC MODEL OUTPUT

50.58% no evidence

Strongest individual response: Adobe Firefly (raw cosine 0.21147). Lowest individual FPR: Adobe Firefly at 13.87%.

Input 2 · `fake_img.png`



Declared provenance	Synthetic (repository reference)
Container format	PNG
Native resolution	512 × 512 px
File size	573 KB
Analysed region	512 × 512 px centre crop
Runtime	18.5 s (cpu)

Cherry-blossom park scene, supplied with the MaskSim reference implementation as its canonical positive example. The repository does not document which generator produced it.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.97528	1.30%	◀ lowest	likely synthetic
Stable Diffusion 2	0.35783	10.65%		no evidence
Stable Diffusion XL	0.00318	55.42%		no evidence
DALL·E 2	0.01486	59.09%		no evidence
DALL·E 3	0.01033	74.46%		no evidence
Midjourney	0.01594	70.97%		no evidence
Adobe Firefly	0.01275	67.21%		no evidence

GENERIC MODEL OUTPUT

7.02% possibly synthetic

Strongest individual response: Stable Diffusion 1 (raw cosine 0.97528). Lowest individual FPR: Stable Diffusion 1 at 1.30%.

Input 3 · `kitten_real.jpg`



Declared provenance	Photograph (EXIF-verified)
Container format	MPO
Native resolution	3336 × 3000 px
File size	3,506 KB
Analysed region	3336 × 3000 px centre crop
Runtime	20.1 s (cpu)

Domestic photograph. The only file in the set that still carries camera metadata: Panasonic DMC-TS3, 23 May 2012.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00046	44.48%		no evidence
Stable Diffusion 2	0.59567	7.37%	◀ lowest	possibly synthetic
Stable Diffusion XL	0.00073	96.60%		no evidence
DALL·E 2	0.01266	60.48%		no evidence
DALL·E 3	0.00761	80.64%		no evidence
Midjourney	0.00414	98.12%		no evidence
Adobe Firefly	0.28064	10.87%		no evidence

GENERIC MODEL OUTPUT

31.89% no evidence

Strongest individual response: Stable Diffusion 2 (raw cosine 0.59567). Lowest individual FPR: Stable Diffusion 2 at 7.37%.

Input 4 · `mountain_real.jpg`



Declared provenance	Photograph (asserted)
Container format	JPEG
Native resolution	2732 × 2732 px
File size	983 KB
Analysed region	2732 × 2732 px centre crop
Runtime	18.4 s (cpu)

Sunset photograph of Ama Dablam, Nepal. Square-cropped and EXIF-stripped, as is normal for stock redistribution.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00014	72.94%		no evidence
Stable Diffusion 2	0.15623	15.95%	◀ lowest	no evidence
Stable Diffusion XL	0.00327	54.19%		no evidence
DALL·E 2	0.00514	68.47%		no evidence
DALL·E 3	0.17366	16.19%		no evidence
Midjourney	0.04911	42.61%		no evidence
Adobe Firefly	0.02485	52.25%		no evidence

GENERIC MODEL OUTPUT

55.22% no evidence

Strongest individual response: DALL·E 3 (raw cosine 0.17366). Lowest individual FPR: Stable Diffusion 2 at 15.95%.

Input 5 · `cat.jpg`



Declared provenance	Unverified
Container format	JPEG
Native resolution	1370 × 1148 px
File size	483 KB
Analysed region	1370 × 1148 px centre crop
Runtime	19.5 s (cpu)

Studio-style cat portrait. Carries no camera metadata and was not marked as a photograph by the operator, unlike the two files named **_real.jpg*. Its provenance could not be established, so it is excluded from the accuracy tally and reported separately.

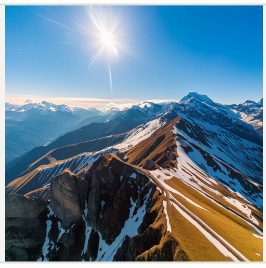
DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00044	45.24%		no evidence
Stable Diffusion 2	0.08273	20.54%		no evidence
Stable Diffusion XL	0.00146	85.25%		no evidence
DALL·E 2	0.07473	43.96%		no evidence
DALL·E 3	0.92716	0.96%	◀ lowest	very likely synthetic
Midjourney	0.03710	47.77%		no evidence
Adobe Firefly	0.82919	2.07%		likely synthetic

GENERIC MODEL OUTPUT

5.25% possibly synthetic

Strongest individual response: DALL·E 3 (raw cosine 0.92716). Lowest individual FPR: DALL·E 3 at 0.96%.

Input 6 · `Adobe_Firefly_…_photography.jpg`



Declared provenance	Synthetic (Adobe Firefly)
Container format	JPEG
Native resolution	960 × 960 px
File size	285 KB
Analysed region	960 × 960 px centre crop
Runtime	19.7 s (cpu)

Landscape image generated with Adobe Firefly.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00081	37.61%		no evidence
Stable Diffusion 2	0.00396	52.54%		no evidence
Stable Diffusion XL	0.00529	34.88%		no evidence
DALL·E 2	0.42483	23.80%		no evidence
DALL·E 3	0.40621	7.52%		possibly synthetic
Midjourney	0.07983	35.08%		no evidence
Adobe Firefly	0.98859	0.28%	◀ lowest	very likely synthetic

GENERIC MODEL OUTPUT

1.66% likely synthetic

Strongest individual response: Adobe Firefly (raw cosine 0.98859). Lowest individual FPR: Adobe Firefly at 0.28%.

Input 7 · `sdxl_text.jpg`



Declared provenance	Synthetic (Stable Diffusion XL)
Container format	JPEG
Native resolution	1024 × 1024 px
File size	368 KB
Analysed region	1024 × 1024 px centre crop
Runtime	19.8 s (cpu)

Scene containing rendered text, generated with Stable Diffusion XL.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.00580	24.22%		no evidence
Stable Diffusion 2	0.58219	7.48%		possibly synthetic
Stable Diffusion XL	1.00000	0.01%	◀ lowest	very likely synthetic
DALL·E 2	0.00279	74.56%		no evidence
DALL·E 3	0.04112	41.08%		no evidence
Midjourney	0.34632	19.03%		no evidence
Adobe Firefly	0.13005	19.87%		no evidence

GENERIC MODEL OUTPUT

0.10% very likely synthetic

Strongest individual response: Stable Diffusion XL (raw cosine 1.00000). Lowest individual FPR: Stable Diffusion XL at 0.01%.

Input 8 · `stable-diffusion-demo-1-1.webp`



Declared provenance	Synthetic (Stable Diffusion 1.x)
Container format	WEBP
Native resolution	1000 × 738 px
File size	94 KB
Analysed region	1000 × 738 px centre crop
Runtime	19.9 s (cpu)

Demonstration image published for Stable Diffusion 1.x.

DETECTOR	RAW COSINE	FPR		INTERPRETATION
Stable Diffusion 1	0.97877	1.21%		likely synthetic
Stable Diffusion 2	0.02351	31.85%		no evidence
Stable Diffusion XL	0.01610	15.22%		no evidence
DALL·E 2	0.00560	67.77%		no evidence
DALL·E 3	0.91257	1.11%	◀ lowest	likely synthetic
Midjourney	0.16788	26.32%		no evidence
Adobe Firefly	0.31991	9.63%		possibly synthetic

GENERIC MODEL OUTPUT

6.03% possibly synthetic

Strongest individual response: Stable Diffusion 1 (raw cosine 0.97877). Lowest individual FPR: DALL·E 3 at 1.11%.

Input 9 · `Astronaut_Riding_a_Horse_(SD3.5).webp`



Declared provenance	Synthetic (Stable Diffusion 3.5)
Container format	WEBP
Native resolution	1024 × 1024 px
File size	86 KB
Analysed region	1024 × 1024 px centre crop
Runtime	20.1 s (cpu)

Generated with Stable Diffusion 3.5 — a generator for which no detector was trained. Included deliberately as an out-of-distribution test.

DETECTOR	RAW COSINE	FPR	INTERPRETATION
Stable Diffusion 1	0.16241	9.82%	possibly synthetic
Stable Diffusion 2	0.92547	3.39% ◀ lowest	likely synthetic
Stable Diffusion XL	0.00674	27.83%	no evidence
DALL·E 2	0.02728	53.59%	no evidence
DALL·E 3	0.65601	3.66%	likely synthetic
Midjourney	0.45761	15.99%	no evidence
Adobe Firefly	0.25794	11.75%	no evidence

GENERIC MODEL OUTPUT

16.77% no evidence

Strongest individual response: Stable Diffusion 2 (raw cosine 0.92547). Lowest individual FPR: Stable Diffusion 2 at 3.39%.

6 Consolidated results

INPUT	DECLARED	BEST FPR	ATTRIBUTED TO	GENERIC FPR	SYSTEM VERDICT	OUTCOME
`real_img.png`	Real	13.87%	Adobe Firefly	50.58%	no evidence	**correct**
`fake_img.png`	Synthetic	1.30%	Stable Diffusion 1	7.02%	possibly synthetic	**correct**
`kitten_real.jpg`	Real	7.37%	Stable Diffusion 2	31.89%	no evidence	**correct**
`mountain_real.jpg`	Real	15.95%	Stable Diffusion 2	55.22%	no evidence	**correct**
`cat.jpg`	Unverified	0.96%	DALL·E 3	5.25%	possibly synthetic	—
`Adobe_Firefly_…_photography.jpg`	Synthetic	0.28%	Adobe Firefly	1.66%	likely synthetic	**correct**
`sdxl_text.jpg`	Synthetic	0.01%	Stable Diffusion XL	0.10%	very likely synthetic	**correct**
`stable-diffusion-demo-1-1.webp`	Synthetic	1.11%	DALL·E 3	6.03%	possibly synthetic	**correct**
`Astronaut_Riding_a_Horse_(SD3.5).webp`	Synthetic	3.39%	Stable Diffusion 2	16.77%	no evidence	**missed**

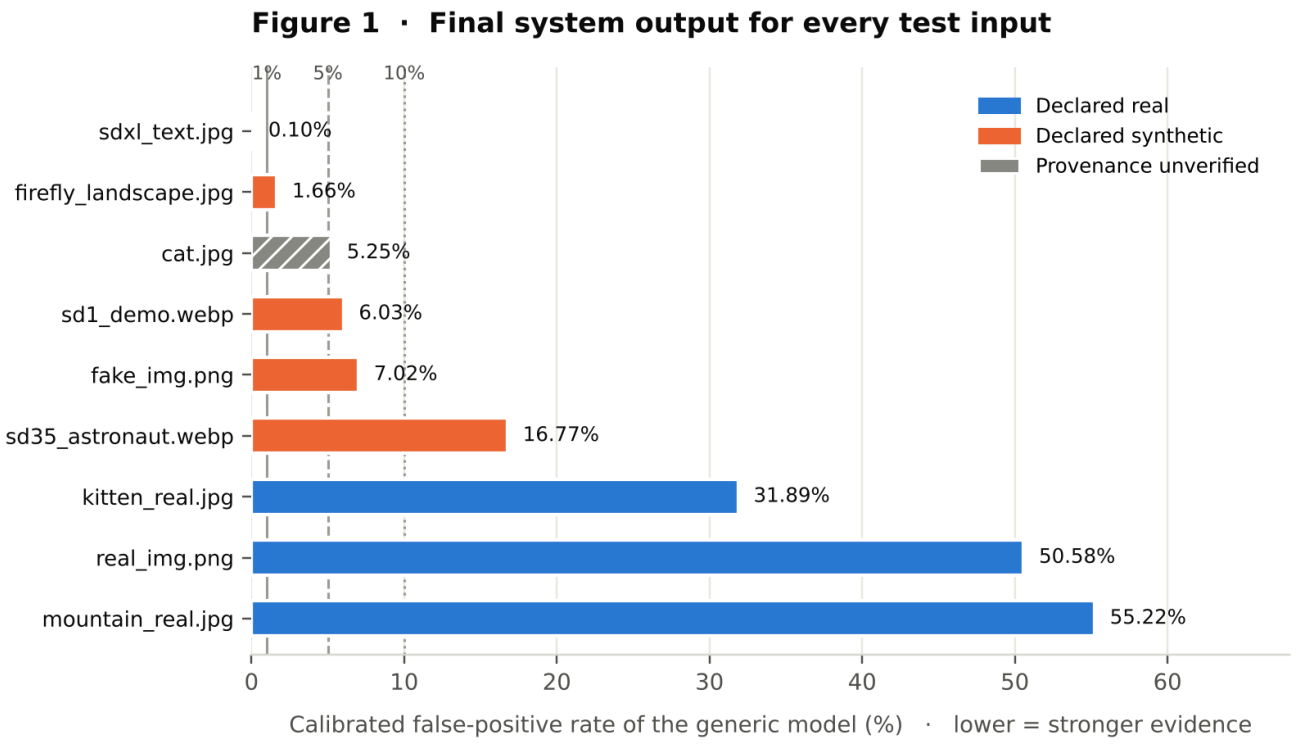


Figure 1. Final generic-model output for every input, sorted by strength of evidence. Dashed guides mark the 1%, 5% and 10% decision thresholds. All three verified photographs fall far to the right of every threshold; four of the five declared synthetic images fall below 10%.

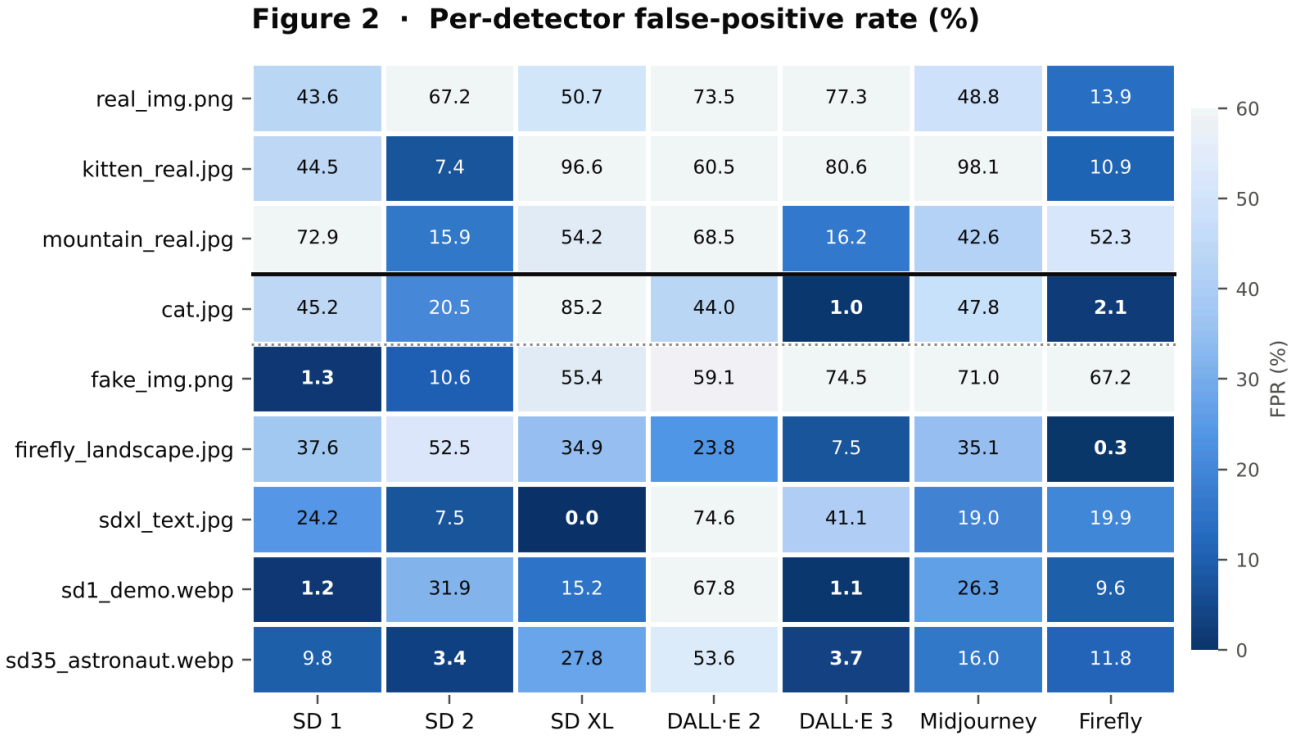


Figure 2. The complete 9 × 7 output matrix: the calibrated false-positive rate of every image against every detector, before the generic model combines them. Darker is stronger evidence; values below 5% are emphasised. The solid rule separates verified photographs from the rest; the dotted rule isolates the unverified file.

7 Outcome of this run

Applying the system's own decision threshold — a generic FPR below 10% counts as a positive detection — to the 8 images whose provenance is declared:

Correctly flagged as synthetic	4 of 5
Missed synthetic images	1 of 5 — `Astronaut_Riding_a_Horse_(SD3.5).webp`
Correctly cleared photographs	3 of 3
False alarms on photographs	0 of 3
Overall	7 of 8 inputs classified in agreement with their declared provenance

The single miss is `Astronaut_Riding_a_Horse_(SD3.5).webp`, generated by Stable Diffusion 3.5. No detector in this system was trained on that generator, and the result — 16.78%, “no evidence” — is the behaviour the method predicts for an unseen architecture rather than a defect in the implementation. It was included in the test set for exactly that reason.

8 Observations

Attribution works where the generator is in the trained set. For `sdxl_text.jpg` the Stable Diffusion XL detector returned a raw cosine of 1.00000 and an FPR of 0.01%, while the other six detectors remained between 7% and 75%. For the Firefly landscape the Firefly detector returned 0.28% against 7.5–53% elsewhere. In both cases the system did not merely detect synthesis — it named the generator, and it named the right one.

The generic model is deliberately more conservative than its best detector. For `fake_img.png` the Stable Diffusion 1 detector reported 1.30%, yet the generic model reported 7.02%. This is not inconsistency: running seven tests and keeping the most extreme result inflates the false-positive rate, so the maximum is re-calibrated against a reference population that was itself passed through the same seven-way maximum. The widening from 1.30% to 7.02% is the multiple-comparisons correction doing its job.

Calibration can reorder near-ties. On `stable-diffusion-demo-1-1.webp` the Stable Diffusion 1 detector produced by far the highest raw similarity (0.97877), which is the correct attribution for that image. After calibration, however, DALL·E 3 ranked marginally lower in FPR (1.11% against 1.21%) and so was selected as the best detector. Raw similarity and calibrated rank are different orderings, and the margin here is within noise.

No photograph was falsely accused. The three verified photographs returned generic FPRs of 31.89%, 50.58% and 55.22% — comfortably inside the region occupied by ordinary camera images. The lowest individual response among them was 7.37%, from the Stable Diffusion 2 detector on `kitten_real.jpg`; the generic model's re-calibration absorbed it, which is precisely the case that correction exists to handle.

The unverified file behaved like a synthetic one. `cat.jpg` drew 0.96% from the DALL·E 3 detector and 2.07% from Firefly, giving a generic FPR of 5.25%. Two independent detectors responding that strongly is not the signature of a photograph. The file carries no camera metadata and was not labelled by the operator, so this report does not assert a ground truth for it and excludes it from the tally in Section 7 — but the measured output is recorded here in full.

9 Reproduction

From the project root, with the virtual environment active:

```
.venv\Scripts\python.exe detect_one_image.py --image_path data/test/sdxl_text.jpg
```

Or, for the interactive version used to produce these figures:

```
.venv\Scripts\python.exe webapp/server.py
# then open http://127.0.0.1:8000 and upload the image
```

The run recorded in this report used the checkpoints and calibration files committed in the repository, unmodified. Results are deterministic: the models are in evaluation mode, no augmentation is applied at inference, and repeating the run reproduces the figures in Section 5 exactly.

10 Limitations and outstanding work

- **Sample size.** Nine images support no claim about accuracy. The numbers in Section 7 describe this run only.
- **Generator coverage.** Detection is confined to the seven generators for which detectors were trained. The Stable Diffusion 3.5 result demonstrates the limit directly.
- **Geometric fragility.** The fingerprint is a set of peaks at fixed frequencies. Resizing by a factor s displaces a peak at bin k by approximately $k(s-1)$, which breaks the match. Resizing was not applied to any input in this test set.
- **No quantitative evaluation.** A receiver-operating-characteristic curve, an area-under-curve figure, a seven-by-seven cross-generator matrix and a robustness sweep against JPEG quality and resize factor have not been computed and are not claimed. These require the full Synthbuster dataset (≈30 GB) and are the natural next step for this project.
- **Provenance.** Ground truth for the test images rests on file names, repository documentation and EXIF records, not on controlled capture. One file could not be verified at all and is reported separately throughout.